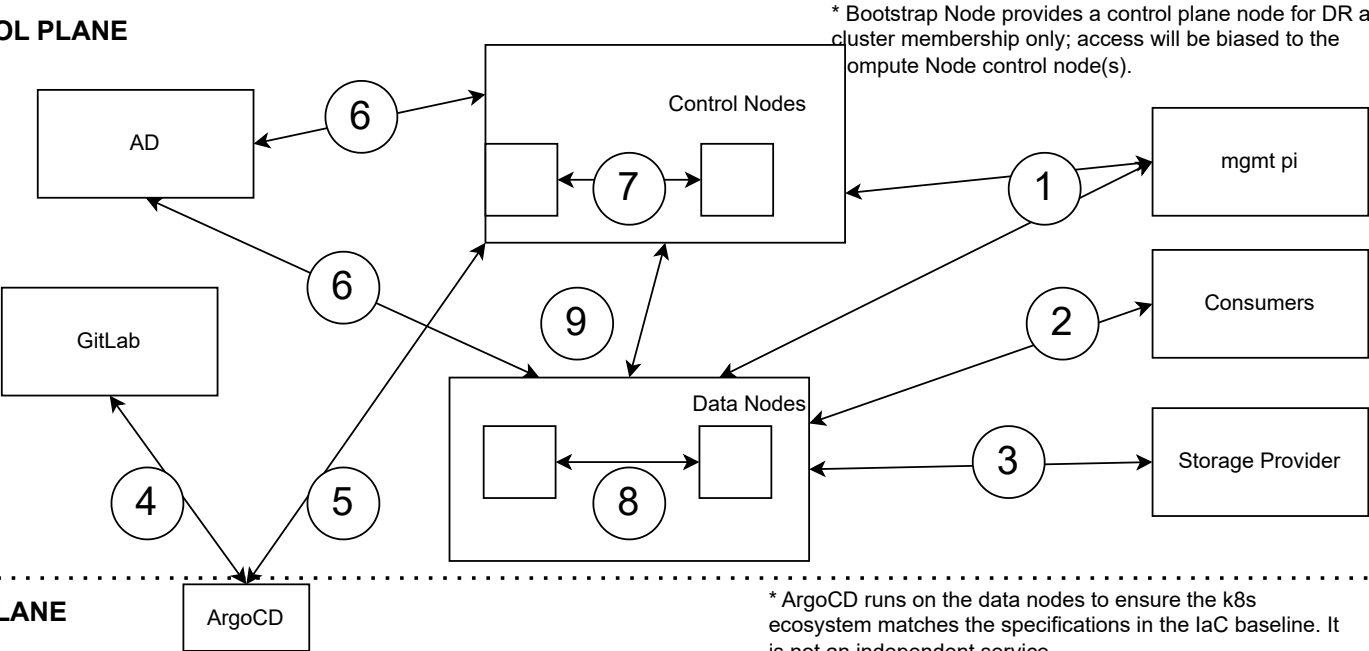


Kubernetes Ecosystem			
Subsystem Code	kb	Component Codes	cont = control plane node data = data plane node

CONTROL PLANE



Data Flows

ID	Port/Proto	Description	Controls
1	50000-50001/TCP 6443/TCP	Node configuration via the TalosCTL API and Kubectl API	Mutual TLS via client certificate issued by Talos-internal CA and stored on mgmt pi
2	Varies	Consumer access to services within the cluster via cluster Ingress Controller or k8s Service	Controls will vary by application/service.
3	3260/TCP	Access to iSCSI volumes for persistent data storage	IQN whitelisting and a separate, non-routed Layer 2 segment.
4	443/TCP	ArgoCD pods to GitLab for GitOps syncs	Narrowly-scoped service account allowing read-only access to Infrastructure Repo in GitLab.
5	6443/TCP	ArgoCD to k8s API server for orchestration actions	Kubernetes Service Account with cluster-internal trust chain. Cannot be used outside the cluster, and can only be used by the service that receives the token.
6	636/TCP 53/UDP	LDAPS and name resolution	Trust relationship with enterprise root CA secures LDAPS. Boundary firewalls prevent name resolution to external sources.
7	2379-2380/TCP 6443/TCP 8472/UDP 4240/TCP-UDP	etcd replication, k8s API server access for internal and management consumer access, and pod networking	Talos-managed PKI validated against an internally generated root CA and mutual TLS.
8	8472/UDP 4240/TCP-UDP 4244/TCP	Pod networking via VXLAN and hubble peering	Talos-managed PKI validated against an internally generated root CA and mutual TLS.
9	10250/TCP 4240/TCP-UDP ICMP 8472/UDP	Kubelet management and Cilium health checks, and VXLAN pod networking	Talos-managed PKI validated against an internally generated root CA and mutual TLS.

Taints		
ID	Name	Result
1	security:NoSchedule	Prevents the node from running pods that don't support the security ecosystem
2	internal:NoSchedule	Prevents the node from running pods that don't support an internal application

RBAC Model			
Name	Project	Resource	Action(s)
Default	All	All	none
OPS-KB-Admins	All	clusters	get, create, update, delete
OPS-KB-Admins	All	projects	get, create, update, delete
OPS-KB-Admins	All	repositories	get, create, update, delete
OPS-KB-Admins	All	certificates	get, create, delete
OPS-KB-Admins	All	gpgkeys	get, create, delete
OPS-KB-Admins	All	logs	get
OPS-KB-Admins	All	exec	create
OPS-KB-<Project>-Developer	<Project>	applications	get, create, update, delete, sync, action, override
OPS-KB-<Project>-Developer	<Project>	applicationsets	get, create, update, delete, sync, action, override
OPS-KB-<Project>-Developer	<Project>	exec	create
OPS-KB-<Project>-Developer	<Project>	logs	get
OPS-KB-<Project>-ReadOnly	<Project>	applications	get
OPS-KB-<Project>-ReadOnly	<Project>	applicationsets	get
OPS-KB-<Project>-ReadOnly	<Project>	logs	get